Mohammad Alfadhel

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Skills

CAD Software: SolidWorks, AutoCAD
Design & Drafting: 3D Modelling, Engineering Drawings, GD&T, Bills of Materials (BOM)
Fabrication & Shop: Basic Machining (lathe & milling machine), Power Tool Operation (drills, impact drivers & angle grinders), Welding (training in progress), Powertrain Assembly, Vehicle Testing, Mechanical Troubleshooting & Repair
Documentation & Data: PDM (Kenesto & 3DX), Technical Report Writing, Microsoft Office (Word, Excel, PowerPoint, Access, Teams), Zoho Workspace, Notion
Safety Training: WHMIS 2015, Worker Health & Safety Awareness, Machine Shop Orientation, Electrical Awareness
Soft Skills: Teamwork, Leadership, Problem-solving, Project management

Experience

Powertrain Team Member May 2026 – Present
University of Waterloo Baja SAE Design Team

- Manufacture and assemble powertrain components, supporting fabrication, installation, and vehicle integration.
- Conduct vehicle testing, including endurance and steering-radius testing, to identify mechanical issues and assess performance.
- Perform post-competition driveline maintenance and diagnostics, including half-shaft removal, reduction box bearing inspection, and CV joint repacking to address drivetrain noise.
- Identified missing fastener torque specifications as the cause of multiple fastener failures during inspection and maintenance; plan to help establish standardized torque specs for the next vehicle.
- Verify vehicle systems for compliance with SAE Baja rules and technical requirements, and support preparation for technical inspection.
- Supported pit crew operations at the OktoBajaFest competition (Colton, NY), contributing to the team's 3rd-place finish in endurance; replaced and bled a ruptured brake line trackside.

IT Network Technician Dec 2024 – Aug 2025
MAF Group

- Streamlined internal workflows and reduced manual data entry time by approximately 20% by designing custom digital tracking forms using MS Access, Zoho, and Excel.
- Tracked expiry dates for 4–5 documents per vehicle across 65 vehicles, keeping records organized and up to date.
- Logged a large volume of purchase invoices in Excel as the record used for tax filing.
- Provided IT support for 8 staff and configured workstations to improve data tracking and internal communication.

Field Project Supervisor (Office Renovation) Mar 2025
MAF Group *Dhahran, Saudi Arabia*

- Supervised a 2-floor office building renovation (3 offices, an employees' lounge, and 2 washrooms), directing an on-site team of up to 8 workers.
- Ensured all renovation tasks met quality and safety standards by verifying workmanship, while handling material procurement and independently resolving on-site logistical challenges to keep the project on schedule.

Projects

Baja Design Projects | *UW Baja SAE Design Team*

- Reverse-modelled the CVT primary currently on the car in SolidWorks to support drivetrain optimization.
- Designed a camera mount in SolidWorks, 3D-printed it, and installed it on the Baja vehicle.

Course Design Projects | *University of Waterloo*

- Created SolidWorks 3D models, engineering drawings with GD&T standards, and bills of materials in the Engineering Graphics and Design course.
- Built a homemade dynamometer with a team in the Introduction to Mechanical Engineering Practice course (ME 100) to test an electric motor, calculating torque from test data in Excel.
- Co-wrote a Feasibility Analysis and Design Evaluation report for a pumpless water gun toy project, and a post-mortem report on a failed gravity-powered toy car.

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science (BASc) in Mechanical Engineering *Expected Graduation: Apr 2031*